

F - Merge Slimes 2

分值: 525525525 分

Problem Statement

There is a sequence AAA of NNN non-negative integers, all initially 000. Process QQQ queries given in order.

For the qqq -th query, you are given integers l_q, r_q, l_q, r_q satisfying $1 \leq l_q \leq r_q \leq N$ and a positive integer a_q . Perform the following in order:

有一个长度为 NNN 的非负整数序列 AAA , 初始时所有元素均为 000。请按顺序处理 QQQ 个查询。对于第 qqq 个查询, 给定满足 $1 \leq l_q \leq r_q \leq N$ 的整数 l_q, r_q 和一个正整数 a_q , 请按顺序执行以下操作:

- Add a_q to each of $A_{l_q}, A_{l_q+1}, \dots, A_{r_q}$.
- Then, letting $M = r_q - l_q + 1$ and $B = (B_1, B_2, \dots, B_M) = (A_{l_q}, A_{l_q+1}, \dots, A_{r_q})$, find the answer to the following problem:
 - 给 $A_{l_q}, A_{l_q+1}, \dots, A_{r_q}$ 中的每个元素都加上 a_q .
 - 然后, 设 $M = r_q - l_q + 1$ 和 $B = (B_1, B_2, \dots, B_M) = (A_{l_q}, A_{l_q+1}, \dots, A_{r_q})$, 求解以下问题的答案:

There are M slimes $1, 2, \dots, M$, where the m -th slime has weight B_m .

Repeat the operation of choosing two slimes and merging them $M - 1$ times.

When slimes of weights x and y are merged, a slime of weight $x + y$ appears and the original two slimes disappear. This incurs a cost of $x \times y$.

Find, modulo 998244353, the minimum possible total cost of the $M - 1$ operations.

有 M 只史莱姆 $1, 2, \dots, M$ ，其中第 m 只史莱姆的重量为 B_m 。重复执行选择两只史莱姆进行合并的操作共 $M - 1$ 次。当重量为 x 和 y 的史莱姆合并时，会产生一只重量为 $x + y$ 的史莱姆，原来的两只史莱姆消失，该操作会产生 $x \times y$ 的代价。求 $M - 1$ 次操作的最小总代价，结果对 998244353 取模。

Note that the changes made on A in each query carry over to subsequent queries.

注意每个查询中对 A 所做的修改会延续到后续的查询中。

Constraints

- $1 \leq N \leq 10^5$
- $1 \leq Q \leq 10^5$
- $1 \leq l_q \leq r_q \leq N$
- $1 \leq a_q \leq 10^9$
- All input values are integers.
- $1 \leq N \leq 10^5$
- $1 \leq Q \leq 10^5$
- $1 \leq l_q \leq r_q \leq N$
- $1 \leq a_q \leq 10^9$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```

NNN QQQ
l1l1l1 r1r1r1 a1a1a1
l2l2l2 r2r2r2 a2a2a2
⋮
lQlQlQ rQrQrQ aQaQaQ

```

Output

Output the answers over QQQ lines in total. The qqq -th line should contain the answer to the qqq -th query.

总共输出 QQQ 行答案，第 qqq 行应包含第 qqq 个查询的答案。

Sample Input 1

```
5 4
1 3 22
3 4 13
5 5 455
1 5 1000000000
```

Sample Output 1

```
1452
455
0
21421644
```

After the first query, $A = (22, 22, 22, 0, 0)$ and $B = (22, 22, 22)$. Merging the first and third slimes first incurs a cost of $22 \times 22 = 484$. Then merging the remaining two slimes incurs a cost of $22 \times 44 = 968$. The total cost is $484 + 968 = 1452$. Moreover, the total cost cannot be made smaller than this.

After the second query, $A = (22, 22, 35, 13, 0)$ and $B = (35, 13)$. The answer is $35 \times 13 = 455$.

第一个查询结束后， $A = (22, 22, 22, 0, 0)$ 且 $B = (22, 22, 22)$ 。先合并第一只和第三只史莱姆会产生 $22 \times 22 = 484$ 的代价，接着合并剩余的两只史莱姆会产生 $22 \times 44 = 968$ 的代价，总代价为 $484 + 968 = 1452$ ，且不存在更小的总代价。第二个查询结束后， $A = (22, 22, 35, 13, 0)$ 且 $B = (35, 13)$ ，答案为 $35 \times 13 = 455$ 。